Applied Advanced Optimisation iRAT 7

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February 27, 2025

Question 1

Explain the two drawbacks of classical convergence analysis.

The constants that determine the rate of convergence are expressed in terms of problem-dependent parameters (e.g., smoothness and strong convexity constants). These parameters are rarely known in practice.

Additionally, the classical analysis relies on bounds that are coordinate-dependent. The problem parameters change with a change of coordinates, even though the actual convergence behavior of Newton's method remains unaffected.

Question 2

Define self-concordance for a function $f : \mathbb{R} \to \mathbb{R}$ and explain why it is coordinate invariant.

A function $f : \mathbb{R} \to \mathbb{R}$ is self-concordant if:

$$|f'''(x)| \leq 2f''(x)^{3/2}, \quad x \in \mathbf{dom}\ f$$

It is coordinate invariant because it only involves second and third derivatives, which remains unchanged under affine transformations.

Question 3

Consider a convex optimisation problem with equality constraints. Explain the method of reduction to unconstrained problems by elimination of the equality constraints.

Given a convex optimization problem:

$$\begin{array}{ll} \min & f(x) \\ s.t. & Ax = b \end{array}$$

The feasible set can be parametrized as:

$$\{x : Ax = b\} = \{z \in \mathbb{R}^{n-p} : Fz + \hat{x}\}.$$

Thus the problem is equivalent to:

$$\min_z \quad f(Fz + \hat{x}).$$

where the solution of the original problem is $x^* = Fz^* + \hat{x}$.

Question 4

Consider a convex optimisation problem with equality constraints. Explain the solution method based on the dual problem.

The dual function of the convex optimisation problem is:

$$g(\nu) = -b^\top \nu + \inf_x (f(x) + \nu^\top Ax) = -b^\top \nu - f^*(-A^\top \nu)$$

where f^* is the conjugate of f. So the dual problem of (1) is

$$\max \quad -b^\top \nu - f^*(-A^\top \nu)$$

Since by assumption there is an optimal point, the problem is strictly feasible, so Slater's condition holds. Therefore strong duality holds, and the dual optimum is attained.

If g is twice differentiable, the optimal solution ν^* can be found solving (5) using the unconstrained minimisation method. The optimal primal solution x^* can then be found by solving $\nabla L(x, \nu^*) = 0$.

Question 5

Explain why the infeasible start Newton's method is not a descent method.

The infeasible satrt Newton's method is not a descent method as it does not always guarantee that the function value decreases monotonically at each step. The derivative:

$$\frac{d}{dt} f(x + t\Delta x)\Big|_{t=0} = \nabla f(x)^\top \Delta x = -\Delta x^\top (\Delta^2 f(x)\Delta x + A^\top w) = -\Delta x^\top \Delta^2 f(x)\Delta x + (Ax - b)^\top w$$

is not necessarily negative unless x is feasible and so $Ax - b = 0$.

Feedback

Q1

(This was the most common error) Some students listed the disadvantages of Newton's method instead of the classical convergence analysis. The actual disadvantages (Section 7.4.1) are: (1) it depends on parameters which are not known and (2) the analysis is coordinate dependent when the method it is analysing is not.

Q2

Some did not explain, or did not do it correctly, why self-concordance is a coordinate invariant property. This can be explained with the argument shown in the third paragraph of Section 7.4.1

Q3

Generally done correctly, but a few students did not provide enough detail or did not reply correctly. The answer is in Section 8.1.1

Q4

As above, done correctly by almost all, answer in Section 8.1.2.

Q5

Some students rushed this point, not explaining that the additional term $(Ax-b)^\top w$ in the derivative is not necessarily negative (paragraph between (11) and (12) in Chapter 8).